



**VU NET**



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An array x of 100 integers is declared as ,

Answer ( Please select your correct option )



int x = [100] ;



int x[100] ;

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com



integer array x = 100 ;



integer x [] = 100 ;



What is the maximum depth of recursive calls a function may make?

Answer ( Please select your correct option )



1



2



n (where n is the argument)



There is no fixed maximum



A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?

Answer ( Please select your correct option )

☐ Arrays

☐ Linked Lists

☒ Binary Search Trees

☐ Stack

Correct answer solved by hadi

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**Made By: Waqar Siddhu**



The new operation in C++ for dynamically allocating memory returns,

Answer ( Please select your correct option )



Size of the memory it has allocated



Pointer to the memory it has allocated



Both size and pointer to the memory it has allocated



Value of the memory it has allocated



In \_\_\_\_\_, a programmer uses two pointers in the node, i.e. one to point to next node and the other to point to the previous node.

Answer ( Please select your correct option )



Linked list

Correct answer solved by hadi  
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Doubly-linked list



Array



Structure

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The expression  $AB+C^*$  is called ?

Answer ( Please select your correct option )



Prefix expression



Postfix expression



Infix expression



Prefix and Infix expression



Consider the function X as under

```
int X (int& Value)
```

```
{  
    return Value;  
}
```

Now **a** and **b** are integers in a calling function. Which one of the following is a valid call to the above function X ?

Answer ( Please select your correct option )



a = X (b) ;



a = X (&b) ;



a = X (\*b) ;



a = X (&\*b) ;

Correct answer solved by hadi  
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Each node in Binary Search Tree has

Answer ( Please select your correct option )

☐ 1 pointer

☒ 2 pointers

☐ 3 pointers

☐ 4 pointers



Suppose  $n$  is the number of nodes in a complete Binary Tree, then maximum steps required for a search operation are

Answer ( Please select your correct option )



☒  $\log_2 (n+1) - 1$

Correct answer solved by hadi  
Cell No:03228043306  
Email: usmanraj20@gmail.com



☐  $\log_2 (n+1)$



☐  $\log_2 (n) - 1$



☐  $\log_2 (n)$

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The **next** field in the last node of a singly-linked list is set to\_\_\_\_\_.

Answer ( Please select your correct option )

☐ NAN

☐ 1

☒ NULL

☐ -1

Correct answer solved by hadi  
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Consider the following function:

```
void test_a(int n)
{
    cout << n << " ";
    if (n>0)
```

Answer ( Please select your correct option )

4 2 0

☐

0 2 4

☐

4 2

☐

2 4

☐



A queue where the dequeue operation does not depend upon FIFO, is called :

Answer ( Please select your correct option )

- ☐ enqueue
- ☐ simple queue
- ☐ stack
- ☒ priority queue

Correct answer solved by hadi  
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$a * (b+c) - d$  is an example of ----- expression.

Answer ( Please select your correct option )

☒ infix

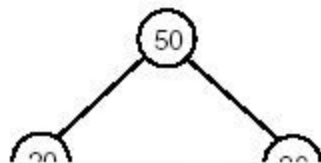
☐ prefix

☐ postfix

☐ allfix



Below is a Binary Search Tree (BST). If we delete the value 50 from the root node, what would be the value in the root of the remaining tree?



Answer ( Please select your correct option )

50



60



70

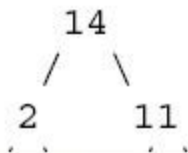


80





Consider the following tree:



Answer ( Please select your correct option )

☐ 2

☐ 4

☐ 6

☐ 9



Which one of the following calling methods does not change the original value of the argument in the calling function?

Answer ( Please select your correct option )



Call by passing the value of the argument

Correct answer solved by hadi

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Call by passing reference of the argument



Call by passing the address of the argument



Call by passing pointer of the argument

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Which of the following statement is NOT true for reference variable ?

Answer ( Please select your correct option )

☐ Once a reference is created, it cannot be later made to reference another object.

☐ References cannot be NULL.

☒ References can be uninitialized.

☐ It is not possible to refer directly to a reference object after it is defined.



Searching an element in an AVL tree takes maximum \_\_\_\_\_ time (where n is number of nodes in AVL tree)

Answer ( Please select your correct option )



$\log_2(n+1)$



$\log_2(n+1) - 1$



$1.44 \log_2 n$



$1.66 \log_2 n$

Correct answer solved by hadi  
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In the post-order traversal of a binary search tree, nodes process as:

Answer ( Please select your correct option )



Left-subtree , Right-subtree , Root



Right-subtree , Root , Left-subtree



Left-subtree , Root , Right-subtree



Right-subtree , Left-subtree , Root



The main use of AVL tree is:

Answer ( Please select your correct option )



Searching of data

Correct answer solved by hadi  
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Storing of data



Insertion of data



Security of data

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Each operator in a postfix expression refers to the previous \_\_\_\_\_ operand(s).

Answer ( Please select your correct option )



One

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Two



Three



Four

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Which one of the following statements is correct?

Answer ( Please select your correct option )



Array size is fixed once it is created.



Link List size is fixed once it is created.



Binary Search Tree size is fixed once it is created



AVL Tree size is fixed once it is created



Linked list always contains elements that can be described as ,

Answer ( Please select your correct option )



Redundant



Recursive



Self-referential.



Bidirectional.

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One difference between a queue and a stack is:



Answer ( Please select your correct option )



Queues require dynamic memory, but stacks do not.



Stacks require dynamic memory, but queues do not.



Queues use two ends of the structure; stacks use only one.



Stacks use two ends of the structure, queues use only one.

Stack and Queue can be implemented using \_\_\_\_\_.

Answer ( Please select your correct option )

☒ Singly Link List

Correct answer solved by hadi  
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☐ Binary Tree

☐ Binary Search Tree

☐ AVL Tree

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New items are added at the \_\_\_\_\_ of the stack.

Answer ( Please select your correct option )

Bottom

☐

Middle

☐

Top



Center

☐



\_\_\_\_\_ is the stack characteristic but \_\_\_\_\_ was implemented because of the size limitation of the array.

Answer ( Please select your correct option )

☐ isFull(),isEmpty()

☐ pop(), push()

☒ isEmpty() , isFull()

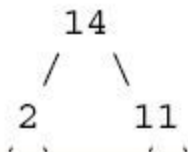
☐ push(),pop()

Correct answer solved by hadi  
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Consider the following tree.



Answer ( Please select your correct option )

☐

2

☐

3

☐

4

☐

5



The difference between a "Binary Tree (BT)" and a "Binary Search Tree(BST)" is that ,

Answer ( Please select your correct option )



A BST has two children per node whereas a BT can have none, one, or two children per node



In BST, nodes are inserted based on the values they contain

Correct answer solved by hadi

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In BT, nodes are inserted based on the values they contain



There is no difference



Which of the following operations returns "most recently entered value" from the stack ?

Answer ( Please select your correct option )



Push



Recent



Top



First

Correct answer solved by hadi  
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Which of the following is correct about AVL Tree?

Answer ( Please select your correct option )



It is identical to BST except height of the left and right subtrees can differ by at least 1.



It is identical to BST except height of the left and right subtrees must differ by at least 1.



It is not identical to BST, its totally different kind of tree.



It is identical to BST except height of the left and right subtrees can differ by at most 1.



What will be postfix expression of the following infix expression?

Infix Expression :  $a+b*c-d$

Answer ( Please select your correct option )



$ab+c*d-$



$abc*+d-$



$abc+*d-$



$abcd+*-$

Correct answer solved by hadi

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In the call by ..... methodology, a copy of the object is passed to the called function.

Answer ( Please select your correct option )



Reference



Value



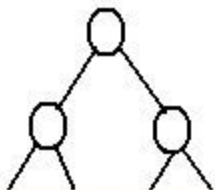
Reference & Value



Copy of the object can not be passed



Consider the following tree, how many levels does it has?



Answer ( Please select your correct option )

One

☐

Two

☐

Three

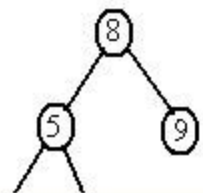
☐

Seven

☐



Consider the following tree, which statement is correct about this tree.



Answer ( Please select your correct option )

☐ 3 and 5 are on the same level

☐ 7 is the parent of 3

☐ 9 is the root of this tree

☐ 5 is on the second level



Recursive call of a function use..... data structure.

Answer ( Please select your correct option )

☐

Linked List

☐

Queue

☒

Stack

☐

Table

Correct answer solved by hadi

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In the statement `int& a= b;`

Answer ( Please select your correct option )

- ☐ a and b pointing to two different memory location
- ☐ a and b are two different names of the same memory location
- ☐ a and b are two different variable names
- ☐ b hold the address of variable a



The variables which are destroyed automatically when a function's execution ends are:

Answer ( Please select your correct option )

☐ Global variables

☒ Local variables defined inside function body

☐ Variables (objects) defined inside function body dynamically

☐ Variables (objects) defined inside function body statically

Correct answer solved by hadi  
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The worst case of searching in binary search tree (BST) is:

Answer ( Please select your correct option )

☒ When the data inserted in BST is sorted

☐ When the height of left sub-tree is greater than right sub-tree

☐ When the height of right sub-tree is greater than left sub-tree

☐ When the tree is balanced



In simple implementation of stack, isFull() method is used due to .....

Answer ( Please select your correct option )



Limitation of array

Correct answer solved by hadi

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Strength of array



Linked list connectivity



Complexity of linked list

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Each node in singly link list has,

Answer ( Please select your correct option )



1 pointer

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2 pointers



3 pointers



4 pointers

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Parameters in function call are passed using,

Answer ( Please select your correct option )



Stack



Queue



Binary Search Tree



AVL Tree



The \_\_\_\_\_ method of list data structure removes the element residing at the current position.

Answer ( Please select your correct option )

Add

☐

Next

☐

Remove

☒

Find

☐

Correct answer solved by hadi  
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Insertion in a linked list can be done at

Answer ( Please select your correct option )



Front only



Back only



Somewhere in middle only



Front, back and somewhere in the middle



An array is a group of ----- memory locations.

Answer ( Please select your correct option )



Scattered



Isolated



Random(non-consecutive)



Consecutive

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Which of the following applications may use a stack?

Answer ( Please select your correct option )



Accessing shared resource



Parentheses balancing program



Buffering messages



Waiting list



Every AVL is \_\_\_\_\_

Answer ( Please select your correct option )



Ternary Tree



Complete Binary Tree



Heap



Binary Search Tree

Correct answer solved by hadi

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Consider the following infix expression:

$$x - y * a + b / c$$

Which of the following is a correct equivalent expression for the above?

Answer ( Please select your correct option )



$$x \ y \ -a \ * \ b \ +c \ /$$



$$x \ *y \ a \ - \ b \ c \ / \ +$$



$$x \ y \ a \ * \ - \ b \ c \ / \ +$$



$$x \ y \ a \ * \ - \ b / \ + \ c$$



Non-recursive calls are faster than which of the following calls?

Answer ( Please select your correct option )

☐ Parameterized

☒ Recursive

☐ Function

☐ Non-Function

Correct answer solved by hadi  
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The balance of a node is the result of "height of left subtree" ..... "height of right subtree".

Answer ( Please select your correct option )



Plus



Minus



Multiply



Divided by



Searching an element in an AVL tree takes maximum \_\_\_\_\_ time (where n is number of nodes in AVL tree)

Answer ( Please select your correct option )



$\log_2(n+1)$



$\log_2(n+1) - 1$



$1.44 \log_2 n$



$1.66 \log_2 n$

Correct answer solved by hadi

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A complete binary tree having “N” nodes consists of ..... Levels.

Answer ( Please select your correct option )

☒  $\log_2 (N+1) - 1$

☐  $\log_2 (N-1) - 1$

☐  $\log_2 (N+1) + 1$

☐  $\log_2 (N-1) + 1$



In the post-order traversal of a binary search tree, nodes process as:

Answer ( Please select your correct option )



Left-subtree , Right-subtree , Root

Correct answer solved by hadi

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Right-subtree , Root , Left-subtree



Left-subtree , Root , Right-subtree



Right-subtree , Left-subtree , Root

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The simplest case in a BST to delete a node is:

Answer ( Please select your correct option )



When the node, that is to be deleted is root node



When the node, that is to be deleted has both left and right child



When the node, that is to be deleted has only one child



When the node, that is to be deleted is a leaf node



If class A defines class B as its friend, then:



Answer ( Please select your correct option )



Class A can access private members of class B



Class B can access only the public members of class A



Class A can access only the public members of class B



Class B can access private members of class A



In the statement `int& a= b;`

Answer ( Please select your correct option )

- ☐ a and b pointing to two different memory location
- ☐ a and b are two different names of the same memory location
- ☐ a and b are two different variable names
- ☐ b hold the address of variable a

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The main use of AVL tree is:



Answer ( Please select your correct option )



Searching of data



Storing of data



Insertion of data



Security of data



In simple implementation of stack, isFull() method is used due to .....

Answer ( Please select your correct option )



Limitation of array

Correct answer solved by hadi

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Strength of array



Linked list connectivity



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Write in postfix form:  $5 * (9 - 7)$

Answer ( Please select your correct option )



$5 (9 7 -) *$



$5 9 * 7 -$



$5 9 7 - *$



$5 (9 7 -)$

Correct answer solved by hadi  
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What will be result of following postfix expression ?

1 2 3 \* + 2 -

Answer ( Please select your correct option )

3



4



(3 + (2 \* 9)) - 6 becomes 3 2 9 \* + 6 -

5



10

